

“Software defined infrastructure is set to make the lives of IT heads easy!”

Intel has recently introduced its Xeon processor E5-2600/1600 v3 product families to address the requirements of diverse workloads and the rapidly evolving needs of data centres. The new processor families include performance increases of up to 3x over the previous generation, world-class energy efficiency and enhanced security. To facilitate the explosive demand for software defined infrastructure (SDI), the processors expose key metrics, through telemetry, which enable the infrastructure to deliver services with the best performance, resilience and optimised total cost of ownership. **Diksha P Gupta from Open Source For You spoke to Ravi Gupta, vertical head-public sector and enterprise, enterprise solutions group, Intel India**, about how SDI's evolution will impact data centres in India and how they will change the IT world. Excerpts:

Q How does Intel plan to be a part of the revolution called software defined infrastructure?

Software defined infrastructure (SDI) is the foundation for cloud computing. The digital services economy requires agility and scale, which demand that all infrastructure resources be programmable and highly configurable. These abilities, coupled with telemetry, analytics and automated actions, allow the data centre to become highly optimised. Intel continues to invest in delivering this vision of an automated data centre, and with the new Xeon E5-2600 v3 product family, the company has introduced key sensors and telemetry that further enhance SDI.

The Intel Xeon processor E5-2600 v3 product family introduces new features that provide greater visibility into the system than ever before. A new cache monitoring feature provides data to enable orchestration tools to intelligently place and rebalance workloads, resulting in faster completion times. This also provides the ability to analyse performance anomalies due to competition for cache in a multi-tenant cloud environment, where there is little visibility into what workloads consumers are running.

The new processors also include platform telemetry sensors and metrics for CPU, memory and I/O utilisation. With the addition of thermal sensors for airflow and outlet temperature, the visibility and control has increased significantly from the prior generation. The processors offer a holistic set of sensors and telemetry for any SDI orchestration solution to more closely monitor, manage and control system utilisation to help maximise data centre efficiency for a lower total cost of ownership.

Q Why do you think that software defined infrastructure is poised to change the current state of IT?

Right now, IT involves a lot of intervention from systems administrators and systems. It needs to be more intelligent, wherein the infrastructure itself takes the decisions on resourcing and orchestration, as it is known as, in the IT language. Software defined infrastructure is the key for that, because once the policies are defined in the software layer, then the software can trigger the hardware to change configurations based on what is needed, who needs more resources, RAM, et al, versus the current state of affairs, wherein the administrator does it manually. With SDI, what we see is that whether it is for storage, networks or computing, the software itself will be intelligent enough to take the decision, or it will be able to trigger the decision in the required manner. This is already happening and we foresee the technology taking over in the next 12-18 months.